

Large Finite Point Sets Have 4 Collinear Points or a 6-Clique

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Abstract

We prove that every finite point set of size at least $10^{11055931}$ has four collinear points or six points that pairwise see each other. This resolves the first open case of the big-line-big-clique conjecture of Kára, Pór, and Wood.

1 Introduction

Given a point set $P \subseteq \mathbb{R}^2$, two distinct points $p, p' \in P$ *see* each other or are *visible* (implicitly, with respect to P) if the only points of P on the line segment $\overline{pp'}$ are p and p' . If instead there is a point $q \in P \setminus \{p, p'\}$ that lies on $\overline{pp'}$, we say that q *blocks* the pair p, p' . The visibility graph of P , denoted by $\mathcal{V}(P)$, has vertex set P and an edge between two distinct points of P whenever they see each other in P . We denote by χ and ω the chromatic number and the clique number, respectively.

Kára, Pór, and Wood conjectured that every sufficiently large finite planar point set has many collinear points or many pairwise visible points [7]. This is commonly referred to as the big-line-big-clique conjecture.

Conjecture 1 (Big-line-big-clique [7]). *For all positive integers k, ℓ , there is an integer $n := n(k, \ell)$ such that every finite point set $P \subseteq \mathbb{R}^2$ of size at least n has ℓ collinear points or $\omega(\mathcal{V}(P)) \geq k$.*

The big-line-big-clique conjecture fails for infinite point sets: there is an infinite point set without four collinear points whose visibility graph is triangle-free [10]. A simple projection argument shows that the conjecture is equivalent if stated in $\mathbb{R}^d$ for any dimension $d \geq 2$ [7].

The conjecture is trivial for $\ell = 3$ (thus $\ell \leq 3$) and every k because if no three points are collinear, then $\mathcal{V}(P)$ is a complete graph. The authors of [Conjecture 1](#) showed the case $k \leq 4$ and every ℓ [7]. The case $k = 5$ and $\ell = 4$ was established [2]. Then [Conjecture 1](#) was proved for $k = 5$ and every ℓ [1], and later the bound in ℓ was improved to $\Theta(\ell^2)$ [3]. Prior to our work, the conjecture was open for all the other pairs (k, ℓ) .

In this note, we show the first open case: $k = 6$ and $\ell = 4$. This marks the first resolution of an open case since 2009, when the arXiv version of [1] appeared.

Theorem 1. *Every finite point set P of size at least $10^{11055931}$ contains 4 collinear points or $\omega(\mathcal{V}(P)) \geq 6$.*

We rely on the following weakening of [Theorem 1](#), due to Hujter and Kisfaludi-Bak.

Theorem 2 ([6]). *Every finite point set P of size at least 2311 contains 4 collinear points or $\chi(\mathcal{V}(P)) \geq 6$.*

Visibility graphs of point sets are not χ -bounded: they can have arbitrarily large chromatic number and clique number 6 [8]. This rules out a general approach to [Conjecture 1](#) solely based on showing its chromatic weakening (where ω is replaced by χ).

Proof outline. Let $P \subseteq \mathbb{R}^2$ be any finite point set without four collinear points. We set $G := \mathcal{V}(P)$, and further assume that $\omega(G) < 6$. We upper-bound $n := |P|$ to prove the contrapositive of [Theorem 1](#).

We first totally order P : $p_1, p_2, \dots, p_n \in P$ such that every interval $p_i, p_{i+1}, \dots, p_j$ induces the same visibility graph as in G . (The order is the left-to-right order on an injective projection of P onto a line.) By the result of Hujter and Kisfaludi-Bak ([Theorem 2](#)), any sufficiently long interval I cannot be made 5-partite (i.e., of chromatic number at most 5) by removing less than a constant fraction of I . Otherwise, a still large remaining subinterval of I contradicts [Theorem 2](#).

As a consequence of a vertex-removal version of the Erdős–Simonovits stability theorem ([Lemma 2](#)), this implies that $G[I] = \mathcal{V}(I)$ has at most $(\frac{2}{5} - \varepsilon_1)|I|^2$ edges for some constant $\varepsilon_1 > 0$. Passing to the complement $B := \overline{G}$, we get that $B[I]$ has at least $c_1|I|^2 - \mathcal{O}(|I|)$ edges for some constant $c_1 > \frac{1}{10}$. The exact value $\frac{1}{10}$ and the strict inequality will matter.

Next, we introduce the following potential, intending to show conflicting upper and lower bounds on it in terms of n ,

$$W := \sum_{p_i p_j \in E(B), i < j} \frac{1}{j-i},$$

summing the reciprocal of the positive *stretch* for every nonvisible pair $p_i, p_j \in P$. Observe that $\sum_{1 \leq i < j \leq n} \frac{1}{j-i} = n \ln n + \mathcal{O}(n)$. Now, because P has no four collinear points, we can show that every unit of potential from a nonvisible pair p_i, p_j forces four units of potential from *visible* pairs; those five units are only used for p_i, p_j and no other nonvisible pair. Therefore, $W \leq \frac{1}{5} n \ln n + \mathcal{O}(n)$.

We now wish to leverage the lower bound on $|E(B[I])|$ for every sufficiently large interval I to get a lower bound on W . Partitioning $[n]$ into intervals of length m , one gets $(\frac{n}{m} - \mathcal{O}(1))(c_1 m^2 - \mathcal{O}(m)) = c_1 n m - \mathcal{O}(n + m^2)$ edges within these intervals. Note that

$$A_m := \sum_{p_i p_j \in E(B), j-i \in [m-1]} \left(1 - \frac{j-i}{m}\right)$$

is the expected number of edges of B within the intervals, thus in particular, $A_m \geq c_1 n m - \mathcal{O}(n + m^2)$. It can be shown that $\sum_{m=j-i+1}^n \frac{2}{m^2-1} \left(1 - \frac{j-i}{m}\right) \leq \frac{1}{j-i}$. By summing this inequality over every nonvisible pair p_i, p_j and swapping the summation order, we get

$$W \geq \sum_{m=2}^n \frac{2}{m^2-1} A_m.$$

With the preceding lower bound on A_m , this gives $W \geq 2c_1 n \sum_{m=2}^n \frac{1}{m} - \mathcal{O}(n) = 2c_1 n \ln n - \mathcal{O}(n)$. We conclude that n cannot be arbitrarily large from the upper bound on W of the previous paragraph, since $2c_1 > \frac{1}{5}$. The section organization follows the paragraphs of the outline.

2 Intervals and lower bound on distance to 5-colorability

We use $[n]$ as a shorthand for $\{1, \dots, n\}$. We use the standard graph-theoretic notation. For any graph H , $V(H)$ and $E(H)$ denote the vertex set and edge set, respectively, of H . If $S \subseteq V(H)$, then $H[S]$ denotes the subgraph of H induced by S , and $H - S := H[V(H) \setminus S]$.

Fix a non-vertical line L such that the orthogonal projection π of P onto L gives $|P|$ distinct points. Let $p_1, \dots, p_n$ enumerate the points of P by the left-to-right order of their projections $\pi(p_1), \dots, \pi(p_n)$. We now call any set of consecutive points $\{p_i, p_{i+1}, \dots, p_j\}$, with $i, j \in [n]$ and $i \leq j$, an *interval* of P .

Observation 1. For every interval I of P , it holds that $\mathcal{V}(I) = G[I]$.

Proof. Two points of I that are visible in P are a fortiori visible in I . Conversely, if two points p_i, p_j of I are blocked by a point $q \in P$, then the projection $\pi(q)$ lies on the line segment $\pi(p_i)\pi(p_j)$. Thus $q \in I$, and p_i and p_j do not see each other in I either. $\square$

Let $n_{\text{HKB}} := 2310$, where the subscript stands for the authors of [Theorem 2](#). The latter theorem can be rephrased as follows: every finite point set with no four collinear points and with a 5-colorable visibility graph has size at most n_{HKB} . We set

$$\varepsilon_0 := \frac{1}{2(n_{\text{HKB}} + 1)}.$$

We show that, due to [Theorem 2](#), the visibility graph of every sufficiently large interval of P cannot be made 5-colorable by removing less than an ε_0 fraction of its vertices.

Lemma 1. *For every interval I of P with $|I| \geq 2n_{\text{HKB}}$, and every $Z \subseteq I$,*

$$\chi(G[I] - Z) \leq 5 \text{ implies that } |Z| \geq \varepsilon_0 |I|.$$

Proof. Indeed, assume that $\chi(G[I] - Z) \leq 5$. Let $z := |Z|$. The set Z cuts the interval I in at most $z + 1$ subintervals. More precisely, $I \setminus Z$ is the disjoint union of at most $z + 1$ intervals $I_1, \dots, I_h$ of P .

For every $j \in [h]$, the interval I_j has no four collinear points, by assumption on P . Furthermore $\mathcal{V}(I_j) = G[I_j]$ by [Observation 1](#). Therefore,

$$\chi(\mathcal{V}(I_j)) = \chi(G[I_j]) \leq \chi(G[I] - Z) \leq 5,$$

where the first inequality holds since $G[I_j]$ is an induced subgraph of $G[I] - Z$. (Note that it is irrelevant that the latter graph is not necessarily a visibility graph.)

The previous paragraph implies, by [Theorem 2](#), that $|I_j| \leq n_{\text{HKB}}$. Consequently,

$$|I| - z \leq hn_{\text{HKB}} \leq (z + 1)n_{\text{HKB}}.$$

Thus,

$$z \geq \frac{|I| - n_{\text{HKB}}}{n_{\text{HKB}} + 1} \geq \frac{|I|}{2} \cdot \frac{1}{n_{\text{HKB}} + 1} = \varepsilon_0 |I|,$$

where the second inequality holds because $|I| \geq 2n_{\text{HKB}}$. □

3 Turán stability and edge density

For any graph H , we use $e(H)$ as a shorthand for $|E(H)|$, i.e., the number of edges of H . We denote by $\text{ex}(m, K_6)$ the maximum number of edges of a K_6 -free (i.e., without a 6-vertex clique) m -vertex graph. We need a vertex-removal version of the Erdős–Simonovits stability theorem [5, 4, 11]. The proof of [9, Lemma 2.3] gives the following explicit form for $r = 5$.

Lemma 2 ([9, Lemma 2.3] for $r = 5$). *For every $\varepsilon \in (0, 1/3750)$, every K_6 -free graph H*

with $m \geq 20$ vertices and more than $\text{ex}(m, K_6) - \varepsilon m^2$ edges

admits a set Z of fewer than $3500\varepsilon m$ vertices such that $\chi(H - Z) \leq 5$.

We get the following useful consequence from the previous two lemmas.

Lemma 3. *Set $\varepsilon_1 := \varepsilon_0/3500$. For every interval I of P with $m := |I| \geq 2n_{\text{HKB}}$,*

$$e(G[I]) \leq \text{ex}(m, K_6) - \varepsilon_1 m^2.$$

Proof. We apply [Lemma 2](#) with $\varepsilon := \varepsilon_1 < 1/3750$ and $H := \mathcal{V}(I) = G[I]$. The conclusion that there is some set $Z \subseteq V(H)$ such that $|Z| < \varepsilon_0 m$ and $\chi(H - Z) \leq 5$ is ruled out by [Lemma 1](#). Since H is K_6 -free (as an induced subgraph of G) and $m = |V(H)| \geq 20$, we have $e(H) \leq \text{ex}(m, K_6) - \varepsilon_1 m^2$. □

We now translate [Lemma 3](#) into a lower bound for $B := \overline{G}$, the complement of G .

Corollary 1. Set $c_1 := \frac{1}{10} + \varepsilon_1$. For every interval I of P with $m := |I| \geq 2n_{\text{HKB}}$,

$$e(B[I]) \geq c_1 m^2 - \frac{m}{2}.$$

Proof. By Lemma 3,

$$e(G[I]) \leq \text{ex}(m, K_6) - \varepsilon_1 m^2 \leq \left(1 - \frac{1}{5}\right) \frac{m^2}{2} - \varepsilon_1 m^2 = \frac{2}{5} m^2 - \varepsilon_1 m^2,$$

where the second inequality is by Turán's theorem [12]. Thus,

$$e(B[I]) \geq \binom{m}{2} - \frac{2}{5} m^2 + \varepsilon_1 m^2 = \frac{m^2}{2} - \frac{2}{5} m^2 + \varepsilon_1 m^2 - \frac{m}{2} = \left(\frac{1}{10} + \varepsilon_1\right) m^2 - \frac{m}{2}. \quad \square$$

The constant $\frac{1}{10}$ and the strict inequality $c_1 > \frac{1}{10}$ will turn out to be crucial.

4 Harmonic upper bound

We define the weight of a pair of points $p_i, p_j \in P$, with $i < j$, by $w(p_i, p_j) := \frac{1}{j-i}$. Note that every weight is positive. We introduce

$$W := \sum_{\substack{p_i p_j \in E(B) \\ i < j}} w(p_i, p_j) = \sum_{\substack{p_i p_j \in E(B) \\ i < j}} \frac{1}{j-i}.$$

Thus, W is the sum of weights of pairs of points that do not see each other in P . The rest of the proof consists of showing conflicting upper and lower bounds on W , resulting in an upper bound on $n = |P|$.

In this section, we show the following upper bound on W . The proof hinges on the absence of four collinear points in P .

Lemma 4. $W \leq \frac{1}{5} n \ln n + \frac{n}{5}$.

Proof. Every nonvisible pair $p_i, p_j \in P$ with $i < j$ admits a *unique* blocker $p_k \in P$, for otherwise there would be four points of P on a single line. We denote by $k(i, j)$ the index of this unique blocker. Furthermore, we have $i < k(i, j) < j$. We charge the weight of (p_i, p_j) to the weights of the three pairs $(p_i, p_{k(i, j)})$, $(p_{k(i, j)}, p_j)$, and (p_i, p_j) . Note that these three pairs cannot be charged by another nonvisible pair: this would again imply the existence of four collinear points in P .

We set $a := k(i, j) - i$ and $b := j - k(i, j)$. Thus, $a + b = j - i$. Therefore,

$$w(p_i, p_{k(i, j)}) + w(p_{k(i, j)}, p_j) + w(p_i, p_j) = \frac{1}{a} + \frac{1}{b} + \frac{1}{a+b}.$$

Since $(a - b)^2 \geq 0$, we have $a^2 + b^2 \geq 2ab$, so $\frac{a}{b} + \frac{b}{a} \geq 2$, as $a, b > 0$. This implies that

$$\frac{a+b}{b} + \frac{a+b}{a} \geq 4, \text{ thus } \frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}.$$

Therefore,

$$w(p_i, p_{k(i, j)}) + w(p_{k(i, j)}, p_j) + w(p_i, p_j) \geq \frac{4}{a+b} + \frac{1}{a+b} = \frac{5}{a+b} = 5w(p_i, p_j).$$

We conclude that

$$W = \sum_{\substack{p_i p_j \in E(B) \\ i < j}} w(p_i, p_j) \leq \frac{1}{5} \sum_{\substack{p_i p_j \in E(B) \\ i < j}} (w(p_i, p_{k(i, j)}) + w(p_{k(i, j)}, p_j) + w(p_i, p_j)) \leq \frac{1}{5} \sum_{1 \leq i < j \leq n} \frac{1}{j-i},$$

where the second inequality holds because the weights $w(p_i, p_{k(i, j)})$, $w(p_{k(i, j)}, p_j)$, $w(p_i, p_j)$ are used only once, by the argument in the first paragraph. Finally,

$$\sum_{1 \leq i < j \leq n} \frac{1}{j-i} \leq n \sum_{1 \leq i \leq n-1} \frac{1}{i} \leq n \ln n + n, \text{ and } W \leq \frac{1}{5} n \ln n + \frac{n}{5}. \quad \square$$

Again, the exact coefficient $\frac{1}{5}$ of $n \ln n$ (and the fact that it is twice $\frac{1}{10}$) will be crucial.

5 Harmonic lower bound

We say that a pair $p_i, p_j \in P$ has *stretch* d if $|i - j| = d$. We first need the following preparatory lemma, which determines the coefficients λ_m for which $\sum_{m>d} \lambda_m \left(1 - \frac{d}{m}\right)$ is equal to $\frac{1}{d}$, i.e., to the weight of a pair p_i, p_j of stretch d .

Lemma 5. *For every positive integer d ,*

$$\sum_{m=d+1}^{+\infty} \frac{2}{m^2-1} \left(1 - \frac{d}{m}\right) = \frac{1}{d}.$$

Proof. First note that $\frac{2}{m^2-1} = \frac{1}{m-1} - \frac{1}{m+1}$. Thus,

$$\sum_{m=d+1}^{+\infty} \frac{2}{m^2-1} \left(1 - \frac{d}{m}\right) = \sum_{m=d+1}^{+\infty} \left(\frac{1}{m-1} - \frac{1}{m+1}\right) - d \left(\sum_{m=d+1}^{+\infty} \frac{1}{m(m-1)} - \sum_{m=d+1}^{+\infty} \frac{1}{m(m+1)}\right).$$

By telescoping,

$$\sum_{m=d+1}^{+\infty} \left(\frac{1}{m-1} - \frac{1}{m+1}\right) = \frac{1}{d} + \frac{1}{d+1}.$$

Moreover, $\frac{1}{m(m-1)} = \frac{1}{m-1} - \frac{1}{m}$, so

$$\sum_{m=d+1}^{+\infty} \frac{1}{m(m-1)} = \sum_{m=d+1}^{+\infty} \left(\frac{1}{m-1} - \frac{1}{m}\right) = \frac{1}{d} \quad \text{and} \quad \sum_{m=d+1}^{+\infty} \frac{1}{m(m+1)} = \frac{1}{d+1}.$$

Therefore,

$$\sum_{m=d+1}^{+\infty} \frac{2}{m^2-1} \left(1 - \frac{d}{m}\right) = \frac{1}{d} + \frac{1}{d+1} - \frac{d}{d} + \frac{d}{d+1} = \frac{1}{d}. \quad \square$$

We can now show the following lower bound on W by leveraging [Corollary 1](#). We set the constant $m_0 := 2n_{\text{HKB}} = 4620$.

Lemma 6. $W \geq 2c_1 n \ln n - 3c_1 \ln(4m_0) n$.

Proof. We have

$$W = \sum_{\substack{p_i p_j \in E(B) \\ i < j}} \frac{1}{j-i} = \sum_{d \geq 1} \sum_{\substack{p_i p_j \in E(B) \\ j-i=d}} \frac{1}{d}.$$

By [Lemma 5](#), and since every term of the series is positive, we have

$$\frac{1}{d} \geq \sum_{m=d+1}^n \frac{2}{m^2-1} \left(1 - \frac{d}{m}\right).$$

Thus,

$$W \geq \sum_{d \geq 1} \sum_{\substack{p_i p_j \in E(B) \\ j-i=d}} \sum_{m=d+1}^n \frac{2}{m^2-1} \left(1 - \frac{d}{m}\right) = \sum_{\substack{p_i p_j \in E(B) \\ i < j}} \sum_{m=j-i+1}^n \frac{2}{m^2-1} \left(1 - \frac{j-i}{m}\right).$$

Interchanging the order of summation, we get

$$W \geq \sum_{m=2}^n \sum_{\substack{p_i p_j \in E(B) \\ j-i \in [m-1]}} \frac{2}{m^2-1} \left(1 - \frac{j-i}{m}\right) = \sum_{m=2}^n \frac{2}{m^2-1} \sum_{\substack{p_i p_j \in E(B) \\ j-i \in [m-1]}} \left(1 - \frac{j-i}{m}\right).$$

Thus

$$W \geq \sum_{m=2}^n \frac{2}{m^2-1} A_m, \quad (1)$$

by setting

$$A_m := \sum_{\substack{p_i p_j \in E(B) \\ j-i \in [m-1]}} \left(1 - \frac{j-i}{m}\right).$$

Our goal is now to lower-bound A_m in some regime of m .

Consider the m ways of partitioning the integers into intervals of size m , intersected with $[n]$. We say that an interval partition *captures* an edge e if both endpoints of e are in the same interval of the partition. Drawing one of the m partitions uniformly at random, the probability that an edge of B of stretch d is captured is $\max(1 - \frac{d}{m}, 0)$. Thus A_m is the expected number of edges of B captured by the partition.

Fix any $m \geq m_0 = 2n_{\text{HKB}}$ with $m \leq n' := \lfloor \frac{n}{4} \rfloor$. We assume that $m_0 \leq n'$, since otherwise the lemma statement trivially holds from the nonnegativity of W . For each partition, there are at least $\frac{n}{m} - 2$ intervals of size m fully contained in $[n]$. Thus, by [Corollary 1](#), any partition captures at least

$$\left(\frac{n}{m} - 2\right) \left(c_1 m^2 - \frac{m}{2}\right).$$

In particular,

$$A_m \geq \left(\frac{n}{m} - 2\right) \left(c_1 m^2 - \frac{m}{2}\right) \geq c_1 m n - 2c_1 m^2 - \frac{n}{2}.$$

Together with [Eq. \(1\)](#), this implies that

$$W \geq \sum_{m=2}^n \frac{2}{m^2-1} A_m \geq \sum_{m=m_0}^{n'} \frac{2}{m^2-1} A_m \geq \sum_{m=m_0}^{n'} \frac{2}{m^2-1} \left(c_1 m n - 2c_1 m^2 - \frac{n}{2}\right).$$

Therefore,

$$W \geq 2c_1 n \sum_{m=m_0}^{n'} \frac{m}{m^2-1} - 4c_1 \sum_{m=m_0}^{n'} \frac{m^2}{m^2-1} - n \sum_{m=m_0}^{n'} \frac{1}{m^2-1}. \quad (2)$$

We observe that

$$\sum_{m=m_0}^{n'} \frac{m}{m^2-1} \geq \ln \frac{n'+1}{m_0} \geq \ln \frac{n}{4m_0}.$$

Moreover,

$$\sum_{m=m_0}^{n'} \frac{m^2}{m^2-1} \leq 2n' \leq \frac{n}{2} \quad \text{and} \quad \sum_{m=m_0}^{n'} \frac{1}{m^2-1} \leq \frac{3}{4}.$$

We therefore conclude from [Eq. \(2\)](#) that

$$W \geq 2c_1 n \ln n - \left(2c_1 \ln(4m_0) + 2c_1 + \frac{3}{4}\right) n \geq 2c_1 n \ln n - 3c_1 \ln(4m_0) n. \quad \square$$

We can now finish the proof of the main result.

Proof of [Theorem 1](#). [Lemmas 4](#) and [6](#) imply that

$$2c_1 n \ln n - 3c_1 \ln(4m_0) n \leq \frac{1}{5} n \ln n + \frac{n}{5}.$$

Therefore,

$$\left(2c_1 - \frac{1}{5}\right) \ln n \leq 3c_1 \ln(4m_0) + \frac{1}{5}.$$

Since the constant $2c_1 - \frac{1}{5}$ is positive (recall that $c_1 > \frac{1}{10}$),

$$\ln n \leq \frac{3c_1 \ln(4m_0) + \frac{1}{5}}{2c_1 - \frac{1}{5}}, \quad \text{thus} \quad n \leq \exp\left(\frac{3c_1 \ln(4m_0) + \frac{1}{5}}{2c_1 - \frac{1}{5}}\right) < 10^{11055931}. \quad \square$$

AI disclosure. After one fruitless attempt and the now customary generic encouragement, a relatively detailed proof of **Theorem 1** was provided by GPT-5.6 Sol Pro after pondering for 222 minutes. The author’s contributions were limited to checking the proof, simplifying some parts, and eventually writing the paper in a way that would give the author (and hopefully other human readers) a more pleasant reading experience.

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